Sofia Salazar-Magaña
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Education

PhD Student in Quantitative and Computational Biosciences 2026 –
Baylor College of Medicine, Houston, TX

Bachelor's degree in Genomic Sciences 2020 – 2024
National Autonomous University of Mexico (UNAM), Querétaro, México.

Research Experience

Data analyst October 2024 – April 2026
Department of Medicine, Section of Genetic Medicine, University of Chicago, Chicago, IL, USA.
PI: Hae Kyung Im, PhD

- Applied a novel lab-developed method TFXcan, to leverage deep learning-based DNA-to-epigenome predictions to investigate TF-binding alterations associated with breast cancer risk.
- Applied the scPrediXcan framework in two projects: (1) training deep-learning models for alternative polyadenylation (APA) prediction, and (2) modeling gene expression across dGTEx developmental contexts.

Undergraduate research assistant July 2023 – October 2024
University of Chicago, Remotely.
PI: Hae Kyung Im, PhD

- Compared the performance of Borzoi and Enformer deep learning models for DNA sequence-based prediction of epigenomic features.
- Provided technical support for lab-developed software enabling TWAS analyses.

Undergraduate research assistant January 2022 – October 2024
International Laboratory for Human Genome Research, National Autonomous University of Mexico, Querétaro, México.
PI: Alejandra Medina-Rivera, PhD

- Analyzed RNA-seq data from the Mexican Lupus Registry (LupusMX) to identify key differentially expressed genes and assembled gene regulatory networks, contributing novel insights into the disease etiology within an understudied population.
- Assembled a bioinformatic pipeline for RNA-seq data analysis, enabling lab-wide workflows.

Contributions

Foguet, C., Gil, L., Xu, Y., **Salazar-Magaña, S.**, Ritchie, S. C., Persyn, E., Im, H. K., Inouye, M., & Lambert, S. A. (2026). OmicsPred as a centralized resource for genetic prediction of multi-omic traits. *Nature Genetics*. https://doi.org/10.1038/s41588-026-02726-4

Zhou, Y., Adeluwa, T., Zhu, L., **Salazar-Magaña, S.**, Sumner, S., Kim, H., Gona, S., Nyasimi, F., Kulkarni, R., Powell, J. E., Madduri, R., Liu, B., Chen, M., & Im, H. K. (2025). scPrediXcan integrates deep learning methods and single-cell data into a cell-type-specific transcriptome-wide association study framework. *Cell Genomics*, *5*(5), 100875. https://doi.org/10.1016/j.xgen.2025.100875

Hernández-Ledesma A. L., Coss-Navarrete, E. L., **Salazar-Magaña, S.**, Ramírez-Espinosa, D., Tinajero-Nieto, L., Torres-Valdez, E., Peña-Ayala, A. H., Félix-Rodriguez, G., Frontana-Vázquez, G., García Sotelo, J. S., Thomas-Chollier, M., Trynka, G., Rosetti, F, Fernandez-Valverde, S. L., Gutiérrez-Arcelus, M., Alpízar-Rodríguez, D. & Medina-Rivera, A. (2026). Gene expression profiling of dendritic cell tolerance dysfunction in women with systemic lupus erythematosus. *Frontiers in Immunology 17*, 1771959. https://doi.org/10.3389/fimmu.2026.1771959

Zhou, Y., Sumner, S., Adeluwa, T., Zhu, L., **Salazar-Magaña, S.**, Kim, H., Gona, S., Nyasimi, F., Kulkarni, R., Powell, J., Madduri, R., Liu, B., Chen, M., & Im, H. K. (2026). Protocol to perform cell-type-specific

transcriptome-wide association study using scPrediXcan framework. *STAR Protocols, 7*(1), 104306. https://doi.org/10.1016/j.xpro.2025.104306

Ogonowski, N. S., Cao, F., Flores-Ocampo, V., **Salazar-Magaña, S**., Seviiri, M., Song, L., Kugelman, J., MacGregor, S., Cuellar-Partida, G., Gharahkhani, P., Im, H. K., Harding, I. H., Yang, J., Nayler, S., Kumar, K. R., Martin, N. G., Diaz-Torres, S., & Rentería, M. E. GWAS Meta-Analysis of Essential Tremor Uncovers 47 Novel Genetic Loci and Biological Pathways Linked to Brain Morphometry of the Ventral Diencephalon and Cerebellum. **Preprint**. https://doi.org/10.1101/2025.11.10.25339873

Conferences

Linking transcription factor binding variation to overall and subtype-specific breast cancer risk variants using a deep learning-based approach. (Poster). *American Society of Human Genetics (ASHG) Annual Conference,* 2025.

Deep learning predicts TF binding variation, illuminating breast and ovarian cancer biology. (Poster). *American Society of Human Genetics (ASHG) Annual Conference,* 2024.

Studying the Systemic Lupus Erythematosus expression signature and contextualizing it in a Mexican cohort. (Talk). *Academic Days at LIIGH Institutional Symposium,* 2024.

Lupus peripheral blood meta-analysis with Recount3 & Recount3-like processed RNA-seq data. (Poster). *Academic Days at LIIGH Institutional Symposium,* 2023.

Lupus peripheral blood meta-analysis with Recount3 & Recount3-like processed RNA-seq data. (Poster). *ISCB-LA SoIBio BioNetMX International Conference on Bioinformatics,* 2022.

Teaching Experience

TA at Deep Learning in Genomics course, *University of Chicago.* Spring ‘26 & ‘25
TA at Quantitative Genomics Training Boot Camp, *Columbia University.* Summer ‘25 & ‘24
Lecturer at bioinformatics institutional courses, *International Laboratory for Human Genome Research.* 2024 & 2023

Certifications & Relevant courses

- **Intro to AI-driven Science on Supercomputers,** *Argonne National Laboratory.*
  Credential ID: a1ff9ac49d6d6754ce667fae36f8ad83f8fe93a5
- **Google’s Data Analytics Professional Certificate**
  Credential ID: KEBDX9FHLDSG
- **Workshop: Analysis of biological and biomedical transcriptomic data using Artificial Intelligence (AI) and Machine Learning (ML) methods with R,** *ISCB-LA SoIBio BioNetMX International Conference on Bioinformatics, 2022.*
- **Workshop: Introduction to the logical modeling of cellular networks, with applications to cancer and immunology**, *International Laboratory for Human Genome Research.*

Skills

- Programming (R, Python, Bash, C++, Git, PHP, SQL, Perl).
- Bioinformatics (bulk and single-cell RNA-seq, genome alignment, epigenomic data analysis e.g ChIP-seq, ATAC-seq).
- Deep learning and acceleration hardware: Pytorch, Tensorflow, CUDA, Intel XPU, MPS.
- Statistics, TWAS, ShinyApp.
- High Performance Computing (SGE, SLURM).

Languages

Spanish (Native) | English (Proficient – TOEFL iBT 118) | French (Basic)